SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: ITA0606 COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)

Assessment Tool Weightage: 40%

QUESTIONS: 15 Total Marks: 25 Annexure A

APPLICATION BASED QUESTIONS

Code of Conduct:

I S.ROHINI / 192424413 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.ROHINI

Annexure A

Short Answer Test

1. A factory has three boxes containing colored balls. Box A contains 6 red, 4 blue, and 2 green balls; Box B contains 5 red, 7 blue, and 3 green balls; and Box C contains 8 red, 2 blue, and 5 green balls. One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C? (5 Marks)

Given:

BOX	RED	BLUE	GREEN	TOTAL BALLS
A	6	4	2	12
B	5	7	3	15
C	8	2	5	15

Since Box B is **twice as likely** to be selected:

$$P(A) = \frac{1}{4},\ P(B) = \frac{1}{2},\ P(C) = \frac{1}{4}$$

Probability of drawing a blue ball:

$$P(\text{Blue} \mid A) = \frac{4}{12} = \frac{1}{3}$$

$$P(\text{Blue} \mid B) = \frac{7}{15}$$

$$P(\text{Blue} \mid C) = \frac{2}{15}$$

Applying Bayes' Theorem

$$P(C \mid \text{Blue}) = \frac{P(\text{Blue} \mid C) \times P(C)}{P(\text{Blue} \mid A)P(A) + P(\text{Blue} \mid B)P(B) + P(\text{Blue} \mid C)P(C)}$$

Substitute the values:

$$= \frac{\left(\frac{2}{15}\right)\left(\frac{1}{4}\right)}{\left(\frac{1}{3}\right)\left(\frac{1}{4}\right) + \left(\frac{7}{15}\right)\left(\frac{1}{2}\right) + \left(\frac{2}{15}\right)\left(\frac{1}{4}\right)}$$

$$= \frac{\frac{1}{30}}{\frac{1}{12} + \frac{7}{30} + \frac{1}{30}}$$

Taking the LCM as **60**:

$$= \frac{\frac{2}{60}}{\frac{5}{60} + \frac{14}{60} + \frac{2}{60}} = \frac{2}{21}$$

Final Answer

$$P(C \mid \text{Blue}) = \frac{2}{21} \approx 0.0952$$

Therefore, the probability that the selected box was Box C is

$$\boxed{\frac{2}{21} \approx 9.52\%}$$

2. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs: (5 Marks)

(a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4 (b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2 (c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0 (d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1 (e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3 (f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6 (g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9 (h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2

Given Data:

EPOCH	TRAINING LOSS	VALIDATION LOSS
1	2.8	2.4
2	2.5	2.2
3	2.2	2.0
4	2.0	2.1
5	1.8	2.3
6	1.6	2.6
7	1.5	2.9
8	1.3	3.2

i) Identify the epoch at which the model begins to overfit.

The model begins to **overfit at Epoch 4**.

Reason: Up to **Epoch 3**, both training loss and validation loss decrease. From **Epoch 4** onwards, the training loss continues to decrease, but the validation loss starts increasing. This indicates that the model is learning the training data too closely and its performance on unseen data is getting worse.

ii) Suggest two effective techniques to reduce overfitting in this model.

- **Early Stopping:** Stop training when the validation loss starts increasing to prevent the model from overfitting.
- **Regularization (L1/L2 or Dropout):** Add regularization techniques to reduce model complexity and improve generalization.

iii) Discuss how overfitting affects the model's generalization ability on unseen data.

Overfitting causes the model to memorize the training data instead of learning general patterns. As a result, it achieves **low training loss** but performs **poorly on unseen or test data**, leading to reduced prediction accuracy and poor generalization.

3. In an industrial process, an AI robot is assigned to segregate the front and back tires automatically. Let the number of tires are 250 (125 front and 125 back tires). (5 Marks)

a) Front identified as Front: 119 b) Front identifies as Back: 06 c) Back identified as Back: 120 d) Back identified as Front: 05 Compute: Accuracy, Precision, Sensitivity, Specificity and F1-Score.

Given:

- Total tires = **250**
- Front tires = **125**
- Back tires = **125**

Confusion Matrix:

ACTUAL/PREDICTED	FRONT	BACK
FRONT	119	6
BACK	5	120

Taking **Front Tire** as the **Positive Class**:

- **True Positive (TP)** = 119
- **False Negative (FN)** = 6
- **False Positive (FP)** = 5
- **True Negative (TN)** = 120

a) Front identified as Front: 119 b) Front identifies as Back: 06 c) Back identified as Back: 120 d) Back identified as Front: 05 Compute: Accuracy, Precision, Sensitivity, Specificity and F1-Score.

1. Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$
$$= \frac{119 + 120}{250} = \frac{239}{250} = 0.956$$

Accuracy = 95.6%

2. Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$
$$= \frac{119}{119 + 5} = \frac{119}{124} = 0.9597$$

Precision = 95.97%

3. Sensitivity (Recall)

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$
$$= \frac{119}{119 + 6} = \frac{119}{125} = 0.952$$

Sensitivity = 95.2%

4. Specificity

$$\text{Specificity} = \frac{TN}{TN + FP}$$
$$= \frac{120}{120 + 5} = \frac{120}{125} = 0.96$$

Specificity = 96.0%

5. F1-Score

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$
$$= \frac{2 \times 0.9597 \times 0.952}{0.9597 + 0.952}$$
$$= 0.9558$$

F1-Score = 95.58%

4. A warehouse has two sections. Section A contains 15 defective items and 35 non defective items, while Section B contains 25 defective items and 25 non-defective items. One section is chosen at random, but Section A is three times as likely to be chosen as Section B. An item selected from the chosen section is found to be defective. (5 Marks)

Given:

Section A:

- Defective items = 15
- Non-defective items = 35
- Total items = 50

Section B:

- Defective items = 25
- Non-defective items = 25
- Total items = 50

Section A is **three times as likely** to be selected as Section B.

Therefore,

$$P(A) = \frac{3}{4},\ P(B) = \frac{1}{4}$$

Probability of selecting a defective item:

$$P(D \mid A) = \frac{15}{50} = \frac{3}{10}$$

$$P(D \mid B) = \frac{25}{50} = \frac{1}{2}$$

Applying Bayes' Theorem

Find the probability that the defective item came from **Section B**.

$$P(B \mid D) = \frac{P(D \mid B) \times P(B)}{P(D \mid A) \times P(A) + P(D \mid B) \times P(B)}$$

Substitute the values:

$$= \frac{\left(\frac{1}{2}\right)\left(\frac{1}{4}\right)}{\left(\frac{3}{10}\right)\left(\frac{3}{4}\right) + \left(\frac{1}{2}\right)\left(\frac{1}{4}\right)}$$

$$= \frac{\frac{1}{8}}{\frac{9}{40} + \frac{5}{40}}$$

$$= \frac{\frac{1}{8}}{\frac{14}{40}} = \frac{\frac{5}{40}}{\frac{14}{40}} = \frac{5}{14}$$

$$P(B \mid D) = \frac{5}{14} \approx 0.3571$$

Final Answer

The probability that the defective item was selected from **Section B** is:

$$\boxed{\frac{5}{14} \approx 0.3571}$$

Therefore, the required probability is approximately 35.71%.

5. A medical image classification system using a deep learning framework combined with K-Nearest Neighbor (KNN) yields TP = 180 and TN = 160. Each class contains 1200 samples, and 25% of the dataset is used as test data. Calculate: Accuracy Precision Sensitivity Specificity. (5 Marks)

Given:

- True Positive (TP) = 180
- True Negative (TN) = 160
- Each class contains 1200 samples
- Total samples = 1200 + 1200 = 2400
- Test data = 25% of 2400 = 600 samples

Total test samples = 600

Known:

- TP = 180
- TN = 160

Therefore,

$$FP + FN = 600 - (180 + 160) = 260$$

Since there are 300 positive and 300 negative test samples:

- Actual Positive = 300
- Actual Negative = 300

Hence,

$$FN = 300 - TP = 300 - 180 = 120$$
$$FP = 300 - TN = 300 - 160 = 140$$

So the confusion matrix values are:

- TP = 180
- TN = 160
- FP = 140
- FN = 120

1. Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{180 + 160}{600} = \frac{340}{600} = 0.5667$$

Accuracy = 56.67%

2. Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$
$$= \frac{180}{180 + 140} = \frac{180}{320} = 0.5625$$

Precision = 56.25%

3. Sensitivity (Recall)

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$
$$= \frac{180}{180 + 120} = \frac{180}{300} = 0.60$$

Sensitivity = 60.00%

4. Specificity

$$\text{Specificity} = \frac{TN}{TN + FP}$$
$$= \frac{160}{160 + 140} = \frac{160}{300} = 0.5333$$

Specificity = 53.33%